Technology Stack

Date	15/07/2026
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack Details

S. No	Architecture Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3 (Jinja2 templates)	Full styling control without third-party UI framework overhead - keeps the form and result pages simple and fast-loading.
2	**Backend / Server-Side**	Python 3.12, Flask	Flask is lightweight and ideal for serving templates and wrapping the trained ML model as a prediction endpoint.
3	**Machine Learning**	Scikit-learn, XGBoost, joblib	Scikit-learn provides Logistic Regression, Decision Tree, and Random Forest baselines plus preprocessing utilities; XGBoost delivers the best-performing tuned model (ROC-AUC 0.717); joblib handles model serialization for deployment.
4	**Data Processing & Visualization**	Pandas, NumPy, Matplotlib, Seaborn	Used for merging and cleaning the two source CSVs, feature engineering, and exploratory data analysis/visualization.
5	**Data Storage**	Flat-file CSV (application_record.csv, credit_record.csv); local .pkl model artifacts	No relational database needed - the dataset is static and small enough for direct file-based loading via Pandas.
6	**Cloud / Hosting / Deployment**	Not applicable in this phase - Flask local development server (127.0.0.1:5000)	Project scope covers local prediction serving; no cloud hosting was required for this phase.
7	**Version Control**	Git & GitHub	Tracks notebook, Flask app, and model artifact changes; used for team collaboration and submission.
8	**Third-Party APIs / Other Tools**	None - Kaggle Notebook used as the primary training environment	No external APIs required; Kaggle provides free compute and hosts the dataset directly for model training.